Birds fly through it, clouds are suspended in it, humans admire and routinely traverse it, light from the Sun and stars flutters through it. But what *is* it? What is it made of? Where does it end? How much of it is there? Where does all that blue come from? If it's a commonplace for all humans, if it typifies our world, surely we should know something about it. What is the sky?

In August 1957, for the first time, a human being rose above the blue and looked around—when David Simons, a retired Air Force officer and a physician, became the highest human in history. Alone, he piloted a balloon to an altitude of over 100,000 feet (30 kilometers) and through his thick windows glimpsed a different sky. Now a professor at the University of California Medical School in Irvine, Dr. Simons recalls it was a dark, deep purple overhead. He had reached the transition region where the blue of ground level is being overtaken by the perfect black of space.

Since Simons' almost forgotten flight, people of many nations have flown above the atmosphere. It is now clear from repeated and direct human (and robotic) experience that in space the daytime sky is black. The Sun shines brightly on your ship. The Earth below you is brilliantly illuminated. But the sky above is black as night.

Here is the memorable description by Yuri Gagarin of what he saw on the first spaceflight of the human species, aboard *Vostok 1*, on April 12, 1961:

> The sky is completely black; and against the background of this black sky the stars appear somewhat brighter and more distinct. The Earth has a very characteristic, very beautiful blue halo, which is seen well when you observe the horizon. There is a smooth color transition from tender blue, to blue, to dark blue and purple, and then to the completely black color of the sky. It is a very beautiful transition.

Clearly, the daylit sky—all that blue—is somehow connected with the air. But as you look across the breakfast table, your companion is not (usually) blue; the color of the sky must be a property not of a little air, but of a great deal. If you look closely at the Earth from space, you see it surrounded by a thin band of blue, as thick as the lower atmosphere; indeed, it *is* the lower atmosphere. At the top of